

PAXSM STUDY FLOWCHART

Four linked tasks | 48 unique participants

RECRUITMENT AND WRITTEN CONSENT

Three separately recruited groups: 12 Task 0 researchers, 24 general participants, and 12 Task 3 researchers

TASK 0 XR Questionnaire Practice Survey

12 XR/HCI researchers | approximately 15 minutes

- Survey current XR questionnaire use and administration
- Survey data-review practice and experienced limitations

TASK 1 PAXSM Feasibility and Response-Process Study

24 general participants | Session 1 | approximately 75 minutes

- PAXSM familiarisation and personal reference
- Traditional versus PAXSM during arithmetic workload
- Careful versus Speeded/Constrained PAXSM responding

Heart rate and eye gaze are collected only in Task 1

TASK 2 Dimension-Linked Target-Selection Study

Same 24 general participants | Session 2 | approximately 60 minutes

- Baseline, then Mental, Temporal, and Physical sub-tasks
- Two Combined sub-tasks: Careful and Speeded/Constrained

Each Combined sub-task includes the difficulty manipulations from the Mental, Temporal, and Physical sub-tasks above

- NASA-TLX and confidence report after every sub-task

PREPARE DE-IDENTIFIED TASK 2 RECORDS

Prepare selected records in Traditional and PAXSM record formats

Task 3 researchers receive prepared records, not the complete Task 2 dataset

TASK 3 Researcher Review of Task 2 Records

Separate 12 XR/HCI researchers | approximately 45-60 minutes

- Review de-identified records in both record formats
- Report decisions, evidence, rationale, and confidence

STUDY COMPLETE